TRANSFORMATIONS · CANONICAL FORMS

How simple the matrix can be made

Choosing a basis means choosing P, and the question is how simple $P^{-1}AP$ can be made. The answer depends entirely on what A satisfies — and the list is ordered from the strongest structure to no structure at all.

5
CASES

DIAGONAL — THE BEST CASE 3

1 **Distinct eigenvalues**

n distinct λ_i

$$D = P^{-1}AP$$

Eigenvectors for distinct eigenvalues are automatically independent, so P is invertible without further checking. Sufficient but not necessary — the row below covers the repeated case, and I diagonalizes trivially with one eigenvalue.

2 **Multiplicities agree**

$m_g(\lambda_i) = m_a(\lambda_i)$ for every i

$$D = P^{-1}AP$$

The necessary and sufficient condition. Columns of P are n independent eigenvectors; the question is only whether that many exist, and the multiplicity comparison answers it eigenvalue by eigenvalue.

3 **Real symmetric**

$A = A^{\mathsf{T}}$ — always succeeds

$$D = P^{\mathsf{T}}AP$$

The spectral theorem guarantees not just diagonalizability but an orthonormal eigenbasis, so $P^{-1} = P^{\mathsf{T}}$ and nothing needs inverting. The only row where the hypothesis is checkable at a glance.

AS CLOSE AS THE MATRIX ALLOWS 2

4 **Defective**

$m_g(\lambda) < m_a(\lambda)$ for some λ

$$J = P^{-1}AP$$

Block diagonal with ones on the superdiagonal. The columns of P are eigenvectors plus generalized eigenvectors, and the number of ones counts exactly how many eigenvectors are missing — the Jordan form measures the failure rather than working around it.

5 **Any complex square matrix**

no hypotheses; U unitary

$T = U^{*}AU$, upper triangular

Schur decomposition. Triangular rather than diagonal, but it exists for every square matrix over $\mathbb{C}$ and U is unitary, which makes it numerically stable where the Jordan form is not — the reason libraries compute Schur and not Jordan.

Every row is the same operation with a different guarantee: conjugate by P and read the result. Similarity preserves eigenvalues, trace, determinant, rank and characteristic polynomial, so nothing about the transformation is lost in any of these — only the coordinates change, and the entries below say how much simplicity that buys.